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Network evolution, entrepreneurial success, and regional development

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Social networks of entrepreneurs are theorized as evolving from an idea generating stage, to a business development stage, and finally to a strategic stage. A model of network evolution that relates changing network activity to the increasing complexity facing the entrepreneur is presented in this paper. A field study of premium wineries, operated by their founding entrepreneurs, is used to (1) assess the impact of social networks in the identification of entrepreneurial opportunity, (2) examine the development and use of networks in establishing the business, and (3) explore the role of interorganizational networks in startup firms. The results suggests that both broad social and interorganizational strategic networks are important to successful startup and ongoing competitive advantage. The results suggest that public policy might be better directed toward the enhancement of this type of network activity, rather than the traditional support provided to encourage business formation.

Keywords: entrepreneurship; network evolution; strategic networks; corporate strategies.

1. Introduction

Attempts to increase entrepreneurial behavior have focused on the development of favorable economic infrastructures (Lewis, 1955), and the identification of individuals with entrepreneurial traits (Rotter, 1966; McClelland, 1965) capable of acting as agents of change (Glade, 1967). Efforts to explain entrepreneurship in terms of entrepreneurs' social networks introduce a research focus that concentrates on relationships, and offers an approach to explaining entrepreneurship in more general terms (Aldrich & Zimmer, 1986; Johannisson, 1987).

Existing research suggests that the composition and structure of an individual's social network can have an impact on the decision to start a business (Aldrich, Reese & Dubini, 1989). Once a firm is started the opportunity identification value of the entrepreneur's social network decreases as its value as a coordinating mechanism increases (Johannisson & Nilsson, 1989; Mazzonis, 1989). The 'technopolis' and research park concept are ways government has recently sought to provide a network substitute to facilitate the coordination of business activities (Quere, 1989; Smilor, Gibson & Kozmetsky, 1989).

The way in which linkages between established entrepreneurial organizations are developed and managed, to enhance the competitive posture of these firms, has been given less attention and has usually focused on large multinational organizations. Bygrave's (1988) study showed that networks provided venture capitalists with access to the most favorable investment opportunities, and suggests a role for interorganizational networks in sustaining entrepreneurial success in an environment that requires the constant generation of new 'deals'.

This paper examines the roles and evolution of networks, and their impact on the initial and ongoing success of the entrepreneurial venture. Network evolution is related to entrepreneurs' businesses' changing needs. The paper proceeds according to the following stages. First, a definitional context is developed that relates entrepreneurial behavior to the opportunity structure of the entrepreneur. Ways by which entrepreneurs can exploit their opportunity structures, using social networks, are then discussed. Then the process by which networks are modified to serve the more immediate needs of the newly established firm is discussed. Lastly, network evolution is tied to the competitive strategy of the firm through the development of interorganizational networks, designed to provide synergistic benefits to all network members. This initial and ongoing process of network development and evolution serves as the basic model of this research.

The remainder of the study is devoted to a discussion of a field study of 29 wineries. Founding entrepreneurs were interviewed to determine how networks impacted on the decision to start their winery and the role they played in the ongoing success of their firm. The results are then interpreted in an historical context to develop public policy prescriptions.

2. Entrepreneurship, networks and opportunity

The ability of the individual to receive and interpret information, in the face of environmental turbulence and uncertainty, is central to entrepreneurial behavior. The entrepreneur's social network is like an 'opportunity set' that can be exploited to provide needed information (Aldrich & Whetten, 1981). The nature and extent of their 'opportunity set' as well as the ability to notice and act on the appropriate entrepreneurial opportunities is what explains entrepreneurial behavior (Kirzner, 1973).

The size and richness of the 'opportunity set' helps explain why some individuals and organizations become aware of entrepreneurial opportunities and others do not. There is also the issue of how the centrality of the individual in an 'opportunity set' affects informational flows, which may result in differences in entrepreneurial success. These network differences may account for differences in entrepreneurial behavior, and suggest that the composition and use of a social network by the entrepreneur needs to be included in studies of the entrepreneurial calculus.

The initial network research focused on how ways for successfully managing change were communicated. McKelvey and Aldrich (1983) saw the initial response as a 'blind choice' that was then proven to be correct more by the environment or chance, rather than from managerial insight. However, the imitation, or communication, of successful responses is less random because entrepreneurs can copy successful variations. This is one explanation for the regularity seen in patterns of human interaction (Berkowitz, 1982), and demonstrates how efficient social networks can benefit the entrepreneur.

The purposeful development of social links, usually referred to as 'action sets' (Granovetter, 1973), acts to supplement the entrepreneurs social 'role set' (Aldrich & Whetten, 1981). These sets combine to define the network, while the strength of the ties between the entrepreneur and others in the network determine the quality and quantity of information received (Thorelli, 1986).

For the entrepreneur it is important to take steps to make the network development process more efficient. This could involve action such as maintaining a central communication position, or seeking business, professional, and social contacts that foster network development (Birley, 1985). MacMillan (1981) found that networks can even

provide some political protection for the entrepreneur from those having a 'vested interest in failure' (Aldrich & Zimmer, 1986). This political protection can be important in ventures requiring more time to produce positive cash flows or where entrepreneurs have included powerful mentors in their networks (Carsrud, Gaglio & Olm, 1986).

The use of networks for ongoing strategic advantage by entrepreneurial firms has received much less attention, although Jarillo (1988: 32) saw network management '... as something that entrepreneurs use purposely to obtain a competitive advantage for their firm'. In this sense he moved beyond the informational benefits of networks to their actual use, or exploitation. This theoretical support for viewing networks in strategic terms has received little empirical investigation.

3. Model of network evolution

The social network is extremely important in the pre-startup stages. A large social network assures the entrepreneur of a larger 'opportunity set' from which to draw both intangible information and tangible resources. The strength of ties between the entrepreneur and others in the social network determines the efficiency and accuracy of communication related to entrepreneurial opportunities. Entrepreneurs with larger social networks have more numerous ties and are more likely to get information that disconfirms their existing notion of how the world works. With a small network the ties could be much stronger and more likely to provide information that is already known, and mutually confirms both parties' view of the status quo (Granovetter, 1973; Nelson, 1989). Additionally, the position of the entrepreneur within the social network affects the speed and timeliness of the information transmitted. Those on the fringes of the network may receive highly distorted or widely shared information that has little entrepreneurial value.

The relationship between social networks and opportunity identification, depicted in Figure 1, is the initial stage or entrepreneurial phase of network evolution. Initially, the social network is important because it is through it that entrepreneurial opportunities are communicated. Entrepreneurs, within the social network, will be more capable of noticing these opportunities and of taking action. Those entrepreneurs with larger social networks

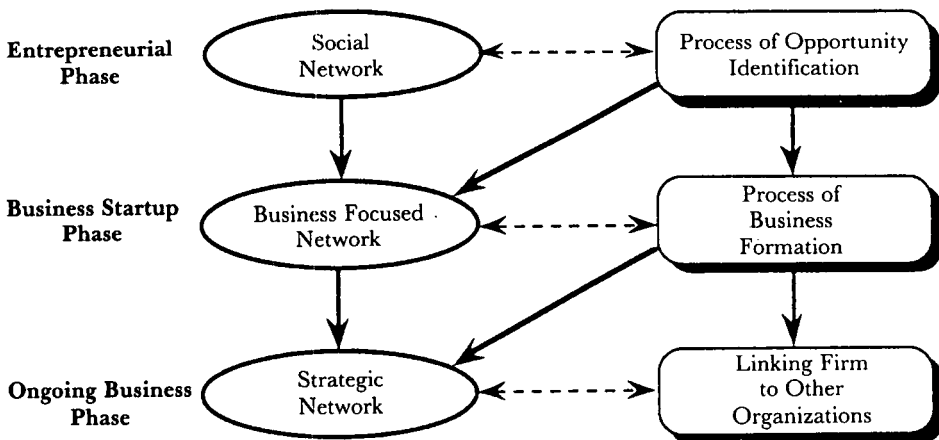


Figure 1. Model of entrepreneurial network evolution.

will have access to more opportunities, which further increases the prospects for noticing entrepreneurial opportunities.

The role of the social network does not conclude with the entrepreneur's decision to start a business because information about future business opportunities can still be communicated through it and some links can be extracted and included in the business focused network needed for the ongoing success of the firm. However, the specific demands associated with the business startup limit the entrepreneur's ability to maintain the preexisting social network because different types of information and tangible resources are now needed.

The model suggests that a more focused network begins to emerge during the startup phase. This business network should reflect links to individuals and organizations that directly serve the more immediate needs of the new business. Thus, the business network is a hybrid that includes both individuals from the preexisting social network, and new individuals and organizations with direct business links. At this stage, startup tasks such as selecting suppliers, raising capital and selecting an organizational structure must be completed. It is this newly developed business focused network that concerns the entrepreneur as new business relationships are consummated. It is evolutionary because it develops gradually and is influenced by the nature of the entrepreneur's social network.

Figure 1 depicts the final stage of network evolution as the development of a strategic network. At this point the entrepreneur has started the firm and is interested in issues such as growth and profit. The entrepreneur becomes aware of the strategic aspects of success and their firm's vulnerability with respect to larger competitors. These strategic networks establish mutual dependence that not only reduce the firm's risk of failure, but also provide advantages not obtainable as an isolated entity. For this reason links with other competitors become important. In some cases these links provide operational information. They can link groups of entrepreneurial firms in ways that cause them to be considered strategic groupings, and as such can generate and share assets such as production knowledge and reputation. In this respect, strategic relationships enhance the competitive posture of all firms in the network.

4. Methods

4.1 Research setting and sample

The study was carried out in Washington state during 1988. This was a period of strong economic growth in the major metropolitan area of the state, which comprises about 50 per cent of the population. However, many areas were depressed economically because of a decline in the primary natural resource based industry with the resulting facility closings and depressed prices for certain agriculture crops. The emergence of a regional wine industry was a bright spot in these regions and provided diversification in an economy that was historically concentrated on a narrow manufacturing base. In addition to its importance to the regional economy, the newness of the industry was an asset for research purposes. In 1980 fewer than a dozen wineries were in business. By the time of this study the state of Washington was only surpassed by the state of California in premium wine production. Of the 78 wineries bonded at the time of this study only one was more than 20 years old. The continuing managerial involvement of the founder-entrepreneurs was critical to this study. Unlike some wine producing regions, the Washington wine industry has not attracted individuals who treat this business as a hobby, although the ability to

generalize our findings to a broader industry setting was a concern, especially because of the possibility that this industry may be characterized by higher levels of social activity. However, the presence of the founder entrepreneurs, together with the smaller size of these wineries, allowed organizational actions and relationships to be reported more accurately.

Interviews were conducted with the founding entrepreneur at 29 wineries. Other managerial personnel and industry experts were also interviewed. In all but one case founding entrepreneurs' interviews were conducted in the wineries. This allowed observational validation with respect to claims about actual and potential capacity, production techniques, and spatial relationships to others in their network. Most of the wineries sampled were started in the 1980s and none was started before 1976. This introduces some age bias but most of the older wineries had changed ownership, or the founding-entrepreneur had abandoned managerial control.

The spatial distribution of wineries is presented in Figure 2. It is presented at this point in the manuscript because in the later discussion references are often made to groups of wineries, and to individual wineries, and these numerical designations on this figure provide some context by which those sets can be interpreted.

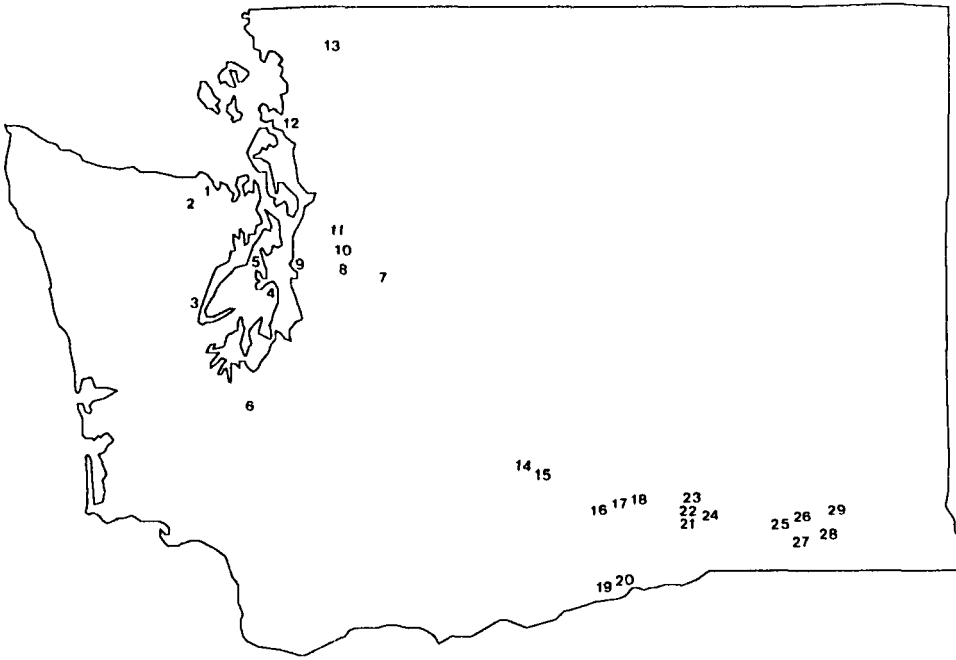


Figure 2. Spatial distribution of wineries.

4.2 Data sources

The founder-entrepreneurs were the primary source for the data used in this research. Initially, a structured interview approach was attempted, which was designed to provide detailed and exhaustive descriptions of each entrepreneur's network. This proved

unworkable as respondents were inclined to provide information in a less structured way, and thus a semistructured interview technique was adopted. The informal nature of the interviews required more time and constant backtracking to ensure that the needed information was collected. The typical interview took about two hours, although in several cases they extended to more than four hours. This allowed respondents to be more spontaneous and permitted the investigators to pursue unexpected avenues of inquiry.

4.3 Measures

Although a qualitative approach was used, the data were systematically collected. Thus, certain issues were addressed in all interviews. These included:

- (a) objective information pertaining to size, capacity, age, organizational structure, product profiles, facilities and technology;
- (b) the size and composition of the social network of the entrepreneur before arriving at the decision to open a winery;
- (c) the identification of individuals and information sources, in their social networks, deemed critical by the entrepreneur in arriving at the decision to start the business;
- (d) the extent that professionals, especially lawyers, accountants and bankers in the entrepreneur's social network, provided information and assistance during the startup phase;
- (e) the extent to which newly established business and social relationships occurred during the startup phase, whom they were with, and the role they played in the success of the new firm;
- (f) an identification of relationships with other wineries or stakeholder organizations including academic institutions specializing in wine making and grape development, industry trade associations, consumer associations interested in wine, government associations established to promote wine, distributors, and downstream retail and restaurant outlets;
- (g) the ways these interorganizational networks were used and the type of information or competitive advantages which they were thought to provide.

4.4 Analysis

In analyzing the data, an attempt was made to reduce certain data so that networks could be illustrated, categorized, and compared. Other data, which seemed meaningful and could not be quantified or classified, were used to provide insights. In many cases, noncategorized incidents enhanced understanding and allowed related issues to be included in subsequent interviews.

Certain modifications became apparent during the data collection effort. These modifications are reflected in the evolution of the model as it is presented in this paper. Although this was not a case of grounded theory development as popularized by Glasser and Strauss (1967), this interplay between theory, data collection, and theory modification is consistent with established views on how data should be used to modify theory (Mintzberg, 1979; Sutton & Callahan, 1987). It should be clear that the analysis here is not intended to be comprehensive, in the sense of explaining statistical variance, but rather it focuses on the process of network development and evolution (Mohr, 1982).

5. Discussion and results

5.1 Social networks

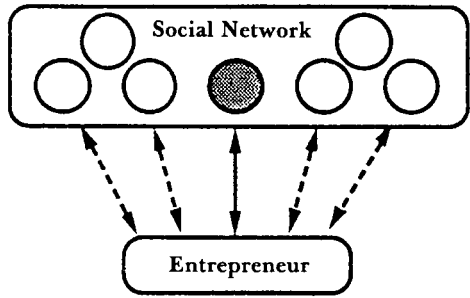
Respondents felt that social networks were important in transmitting information about their entrepreneurial opportunity. However, the composition and size of their networks exhibited considerable variation. In general, the data suggest two continua along which the social networks of entrepreneurs can be categorized, size and degree of wine relatedness. Thus, respondents were categorized as having social networks that were (1) large and unrelated to wine, (2) large and focused on wine, (3) small and focused on wine, and (4) small and unrelated to wine. No respondent was found to have social networks that could be categorized as small and unrelated to wine.

5.1.1 Large and unrelated: Seven entrepreneurs (3, 7, 13, 14, 18, 23 & 29) had extremely large social networks, which were not specifically tied to the wine industry. Because of the extensive nature of their social networks, these entrepreneurs indicated that they had received information relative to a number of entrepreneurial opportunities. In all cases, they had successfully operated other businesses before deciding to open a winery. Figure 3(a) depicts how their network acted to bring the appropriate information to their attention. Their large social network produced a diverse range of opportunity sets, one of which, symbolized by the darkened circle, was related to wine. Entrepreneurs with this type of network are very likely to start a business, but the fact that they ended up in the wine business should be considered more probabilistic. Their social network acted to energize the entrepreneurial process on multiple fronts. This is indicated by the large number of arrows emanating from the social network, each of which could be interpreted as offering business opportunities in a different area.

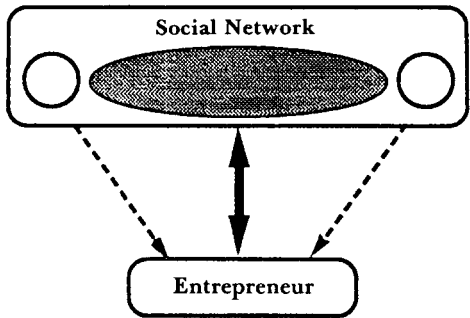
These entrepreneurs stressed the value of social contacts. In four cases startup was the byproduct of a friend's suggestion or business partnership offer. In another case an entrepreneur had been given a set of characteristics required for a successful winery, and then set off in search for the proper site. These entrepreneurs acted quickly and in one case the entrepreneur actually began building the winery the next day. These entrepreneurs had some interest in wine as consumers, but they did not engage in a conscious search directed exclusively toward starting a winery. In addition, the presence of a rich and varied 'opportunity set' allowed the entrepreneurs to compare and contrast alternatives, allowing for a more critical analysis of the wine business prior to making extensive personal contacts.

5.1.2 Large and related: This group of ten entrepreneurs (2, 9, 10, 16, 21, 22, 25, 26, 27 & 28) had extremely large social networks before starting their winery, but they had long-standing desires to start a winery and their social network was already a byproduct of this interest. The social network of these entrepreneurs was as large as those with unrelated social networks, but the wine-related opportunity set was much larger than any other opportunity set in their social network (see Figure 3(b)). Entrepreneurs indicated that their networks were capable of providing information about other types of businesses but that they were more receptive to information related to wine. For this reason a one way informational flow has been depicted between the nonwine-related opportunity set and the entrepreneur. The bold arrow between the entrepreneur and the wine-related opportunity set depicts a wider, and more interactive, breadth of information flow. Entrepreneurs in this group seemed more likely to start a winery, but differed from entrepreneurs with large

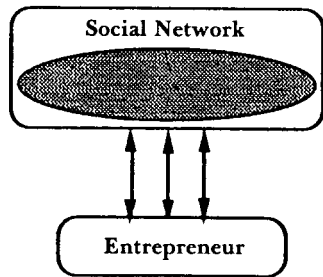
(a) Large and unrelated social network, (b) Large and related social network, (c) Small and focused social network.



(a)



(b)



(c)

Legend:

- Potential entrepreneurial opportunity
- Realized entrepreneurial opportunity

Figure 3. Types of social networks.

and unrelated networks because they were much less likely to start another type of business, and tested their notions of a successful winery against other wineries rather than against other types of business.

Their desire to enter the wine business was deep seated, and it was this desire that affected their receptivity. In many cases, they constructed their social activities around wine and their social network was more restricted in range than the first group of entrepreneurs. Many entrepreneurs in this group had been hobby wine makers previously, had relatives or friends already in the wine business, or lived near wine regions. Several spoke of their desire always to open their own winery.

5.1.3 Small and focused: Twelve entrepreneurs (1, 4, 5, 6, 8, 11, 12, 15, 17, 19, 20 & 24) fell into this category. Their social networks were small, but the proportion of their network that was wine related was large. The network structure characterizing this group is depicted in Figure 3(c). These entrepreneurs had smaller social networks than the other groups of entrepreneurs but the 'opportunity set' devoted to wine comprised almost their entire network. These entrepreneurs tended to minimize contacts with others, spent more time and energy on wine related informational searches, and conducted their search in a more conscious but limited way. Entrepreneurs with this type of network are more likely to end up in the wine business, but it takes longer than for those with larger or broader social networks, because they get more involved in collecting detailed operational information. Thus, the logic behind the multiple direct arrows between the wine-related opportunity set and the entrepreneur in Figure 3(c). Prior to startup, other aspects of their life such as career choice and travel could be viewed as a means to collect capital, knowledge and experience on wine making. Almost all other activities were subservient to wine related activities.

In some sense, these entrepreneurs appear to appraise the industry more as a business than as an entrepreneurial opportunity. One entrepreneur even waited until he could retire to achieve his goal. Another spoke of having wanted to start a winery since his youth. Several linked their business goal to religious convictions, while others saw it as part of some ideal life style. Three entrepreneurs in this group actually moved to the area because they had made a determination that that would be the best place for their operation. As suggested above, the narrowness and intensity of these social networks were not efficient for generating disconfirming information or for developing broader models of business enterprises.

5.2 Business directed network

The development of a business-focused network was not a high priority for the entrepreneurs in this study. They used their social network less than expected, especially with respect to identifying professionals such as accountants, lawyers, and bankers. In fact, the addition of individuals to their network, capable of providing business assistance to the firm during its startup phase was not a high priority. Professionals were seldom consulted during the startup or during the initial stages of business operation. Even the complicated process of bonding their winery, at both the Federal and State level, was usually handled without the assistance of attorneys. Accountants tended not to be consulted, and bankers were 'emphatically' omitted from the information search of most entrepreneurs.

Some entrepreneurs did use their social networks as a source of labor, especially during periods of high labor demand. In one case this included friends of an offspring, and in

another residents of the local retirement community. Later, as the wineries chose legally to incorporate, social networks were occasionally tapped for legal assistance. Accountants were sought out later, but only for the purpose of filing taxes and they seemed less likely than lawyers to be byproducts of the social network.

Entrepreneurs were more uniform in the methods used to identify sources of production, equipment and grapes, or in identifying channels of distribution. They made conscious efforts to make contacts with other wineries to obtain needed information. These between winery contacts became very important to the development of a strategic network. Subscribing to magazines related to wine making such as *Wines and Vines*, membership in industry associations, and initial contacts with universities with grape growing and wine making expertise seemed to characterize this period. Entrepreneurs in the more recently founded wineries (2, 19, 24, 27 & 29) felt these activities were appropriate but that the benefits would flow much later, and seemed unsure as to the form these benefits would take.

No consistent set of patterns emerged that distinguished the firms with respect to their business focus. Those with larger social networks did not appear to have substantially larger business networks, although some entrepreneurs appeared to be making better use of their existing social network. In one case the existing network was tapped as a source of capital and for ongoing managerial talent. In another case the social network of the entrepreneur was actually expanding and being developed, and becoming a tool for distribution. One winery had an annual grape stomping party, which attracted over 1,000 people, that had a positive impact on retail sales at the winery. Other entrepreneurs seemed to be evolving their networks more narrowly, so that they now reflected their totally directed interests in wine. This focused perspective also had certain benefits, especially with respect to the diffusion of useful information between wineries.

What seems apparent was the fact that these entrepreneurs felt the actual business startup was the easiest part of the process. Prior business experience, and the fact that they felt they had a clear grasp of the manufacturing process, led many to downplay the value of business expertise. One entrepreneur even joked that he 'manufactured' the numbers in his business plan 'because bankers are only concerned with how it looks, not if the numbers are accurate'.

5.3 Strategic networks

Recognizing that links with other organizations could be exploited was a key element in the final stage of network evolution. Strategic networks began to emerge after the details of business startup were completed. The marketing aspects of these networks were the most visible, but they also took other forms and linked wineries in ways that enhanced their posture *vis-à-vis* wines produced in other regions. There was a similarity of approach with respect to interorganizational linkages, which allows groups of wineries to be categorized broadly as structurally equivalent. For instance, centrality and density appeared to be more important to some firms, because they see their success directly related to the industry as a whole. These firms consciously and purposely established and managed these interorganizational relations, just as directly and vigorously as they managed their own organizations.

Respondents identified links with other wineries, grape growers, and with industry associations as providing certain strategic advantages. Entrepreneurs attempted to position their firm, *vis-à-vis* the network of all organizations, in a manner that allowed them

to emphasize certain factors. Direct, interorganizational cooperation was the most common theme.

This interorganizational cooperation facilitated competition with larger firms and wines from more established regions. Passively, this occurred as the number of wineries increased and an infrastructure of wine-related businesses emerged, which reduced the transaction costs for producers. The more active phases of this cooperation relate to production and marketing.

On the production side, the transfer of information with respect to managerial techniques and information was considered important to regional success. Because this product tends to have a regional reputation, higher average levels of quality were seen as important to the success of all producers. Respondents saw their success as linked to that of the industry, although entrepreneurs who had been in the business longer placed a higher value on the benefits that flowed from a strongly linked and cohesive industry. The level of interfirm cooperation was higher in wineries located in agriculture regions (14 through 29), where cooperation had historically been viewed as enhancing competition.

On the marketing side, cooperation was much more visible. For instance, all the wines from the region are grouped together on retailer's shelves. There were also common efforts to attract customers to the actual winery, where the margin on sales was higher. An annual hot air balloon festival was used as a focal point by four wineries (25, 26, 27 & 28) to attract tourists to their wineries. A common publicity brochure was used by two sets of wineries ([1, 2, 3, 4, & 5] & [14, 15, 16, 17 & 18]) because owners recognized that buyers are unlikely to expend the effort to visit a single winery. Wine tastings and participation in regional wine festivals were also seen as a way to achieve mutual benefits.

There was less emphasis put on the use of networks as a means to maintain channel power. While many entrepreneurs selected their distributor based on a competitor's recommendation, there was little evidence of using their alliance to maintain supplier power. However, in one case a group of wineries agreed to a common distribution schedule, from a single bonded location, which they felt encouraged the distributor to be more energetic in generating sales for their products.

The formalization of alliances, through the formation of umbrella organizations, was a product of the earlier informal cooperation. The Washington Wine Institute was established as the industry association to monitor legislation affecting the industry, although it eventually became engaged in promotional efforts, especially in other states. The State Wine Commission was established as a way to institutionalize wine promotion, by collecting a small tax on each bottle of wine sold. Several smaller associations were founded to promote wine produced in their specific region.

These umbrella groups were valued less by smaller wineries. However, entrepreneurs from smaller wineries, with larger social networks, saw the activities of these industry associations as having value in the long term. Several of these entrepreneurs were on the State Wine Commission and felt that the agenda should be moved toward setting policies and marketing activities designed to enhance the posture of smaller wineries, which involved local rather than national publicity. The annual production of an art poster with the labels of all the wineries was seen as one such effort, because it served to inform consumers of the full spectrum of products available.

Links to grape growers took different forms. In the eastern part of the state, several entrepreneurs (14, 15, 18, 19, 21, 23 & 24) had actually been grape growers before starting their winery, and saw this as a sort of vertical integration. Another group of wineries (5, 6 & 13) grew specialized varieties that grew best in the western part of the state. This latter group was active in sharing both information and root stock. Another link between

wineries and growers related to product differentiation. Three wineries (11, 25 & 28) had successfully differentiated their product and captured the high end of the regional market. Because of the newness of the industry, grape stock was limited and each of these wineries put a very high value, and spent lots of time, establishing links to growers.

While these wineries were linking themselves to the best growers, they were also attempting to link to each other and limit further entry at the 'super premium' end of the market. All were active in telling the story of a foreign monarch who had sent his buyer to buy a supply of these three wines. Other wineries were enhancing their reputation by linking their wine to one or more of these 'super premium' wines.

6. Implications for policy makers and conclusions

While this research has focused on private sector dynamics, some public policy implications can be deduced from the results. Governmental efforts to encourage commercial activity can benefit from using the existing knowledge base to develop entrepreneurial programs that foster network development. Concern has generally focused on regional development of start up companies and on industries that are, or have the potential to become, economically disadvantaged. This research, when interpreted in the context of existing research by business historians and development economists, suggests that the nature and extent of networks can substantially influence long-term success of new ventures.

In attempting to explain why entrepreneurship emerges, business historians have taken a more micro-approach to the explanation of the role of social relationships. For instance, greater amounts of industrial activity in Northern than in Southern Italy, have been attributed to social flexibility and movement, which allowed greater communication between classes (Clough & Livi, 1956). This same type of social communication was found to be important in explaining why some blacks became entrepreneurs and others did not in the pre-civil war United States. Blacks with white fathers were found to be much more likely to become free and entrepreneurs, and in most cases this is explained by the social links to their white father (Schweninger, 1989). More interesting are social links that affected actual business decisions. Social links between Detroit's upper class resulted in most early car manufacturers building cars for the 'country club' set, while the mass market was ignored. Henry Ford, who abhorred Detroit's upper class, was the first to develop a product for the vast middle-segment market (Davis, 1988).

In other cases the social values of the group, such as with Quakers in eighteenth-century England, were found to be relevant factors to commercial success. The Quaker view that commercial fairness required the setting of fixed prices and the marking of products with these prices found favor with consumers, and led to their commercial success as shopkeepers (Emden, 1940). Like the wine industry, their shared values and experience also led to the development of specific industries, where it was possible to implement these values in a commercial setting (e.g. Smith, Child & Rowlinson). Firms such as J. S. Fry and Sons, H. I. Rowntree & Co., and Cadbury Brothers emerged from the numerous Quaker owned cocoa shops, and the sharing of information about how to remove cocoa butter from chocolate helped foster their domination of this industry (Dellheim, 1987; Raistrick, 1950). A general thrust of this research is that social linkages were relevant factors to commercial success.

On the development side the focus has been on the macro-level economy (e.g., Alexander, 1960; Ashkenazi, 1988; Robbins, 1985). Social conditions such as the ability of

individuals to emerge from the group (Hirschman, 1958) or the family (Habakkuk, 1955; Nafzinger, 1969), a more favorable political environment (Yamamura, 1973), or the ability of a small group of concentrated firms to create a market (Neff, 1978) have been identified as important.

The general theme suggested by the existing research, when seen in the context of this research, is to develop action plans that duplicate historical successes. For instance, the first thing to determine is what are the core values or factors that are the major foci in individuals' existing social networks. Attempts to develop a wine industry, no matter how favorable the economic or geographic conditions, will not meet with success if the indigenous population has shared prohibitionist values.

In other cases, where people have either large and related (Figure 3(b)) or small and focused (Figure 3(c)) networks, attempts at economic development should be made in the context of these foci. Thus, vertical integration, as opposed to unrelated business ventures, may offer greater promise of success. The identification of existing foci should be one of the first concerns of any entrepreneurship development program.

The existence of large and unrelated social networks (Figure 3(a)) provides the greatest potential for broad based economic development, but it may be here that the conditions for economic development become more important. For instance, why did wineries develop in larger numbers in the state of Washington, rather than in other regions having similar climatic and social conditions? Obviously, the existence of certain resources is a necessary condition, but the answer appears to be in the mix of network types, and the evolutionary development of these networks. In Washington state the strong agriculture base produced many entrepreneurs with networks with access to information on grapes and wine. The switch from one agriculture to another is easier than starting an unrelated business. Environmental factors, such as the geographic proximity to the well-developed Californian wine industry and a strong local market for premium wines, were also important. This combination of favorable environmental factors and the existence of focused-network entrepreneurs encouraged the entrepreneurs with large and diverse social networks to select wine making from the portfolio of options in their opportunity set. This latter group of entrepreneurs built some of the largest wineries, and had greater levels of business and marketing experience, which was critical to the ongoing development of the industry.

In addition to providing the information needed to exploit entrepreneurial opportunities, social networks appeared to play an important role in network evolution. For individuals with entrepreneurial capabilities, a broad diversified social network is the most efficient way to become aware of opportunities. Focused networks can lead entrepreneurs to start a specific type of business, but this may occur even if this business does not represent an entrepreneurial opportunity. The Washington state experience may just be a case of 'being in the right place at the right time' for these focused network entrepreneurs.

The need for the development and use of business focused networks was limited. These entrepreneurs seemed to feel that this was the routine aspect, and that identifying opportunities, and then strategically exploiting them was far more important. This may be a reason why government efforts at creating business networks through science parks and technopolises have not met with high levels of success. Since this does not appear to be what differentiates commercial success, it may be appropriate to reduce some of the governmental support that has traditionally focused on these services.

The development of strategic networks, based on interorganizational cooperation was the most interesting finding. This implies that networks should be changed and modified to meet the evolving needs of the organization. It also suggests that organizational links should be stretched to include not only buyers and suppliers but also competitors, because these links can also provide competitive advantage.

The role of government in developing policies which encourage network evolution, from social to strategic, is more complex. Typically, governmental and educational efforts have focused on the technical aspects of the startup. This study suggests this is the least critical stage. Education efforts directed toward building and exploiting social networks, and governmental efforts designed to foster ongoing linkages with established firms would be more likely to enhance regional economic development.

The research also suggests that further theoretical refinements are needed, so that the translation process between network development and use and firm advantage, are more carefully specified. More detailed descriptions of social networks, empirical measurement of linkages, and the identification and validation of strategic networks as a way to achieve competitive advantage are needed to further our knowledge in this area.

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